

3-3 Measures of Position

Standard Scores

A **z score** or **standard score** for a value is obtained by subtracting the mean from the value and dividing the result by the standard deviation. The symbol for a standard score is z . The formula is

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

For samples, the formula is

$$z = \frac{X - \bar{X}}{s}$$

For populations, the formula is

$$z = \frac{X - \mu}{\sigma}$$

The z score represents the number of standard deviations that a data value falls above or below the mean.

EXAMPLE 3-27 Test Scores

A student scored 65 on a calculus test that had a mean of 50 and a standard deviation of 10; she scored 30 on a history test with a mean of 25 and a standard deviation of 5. Compare her relative positions on the two tests.

SOLUTION

First, find the z scores. For calculus the z score is

$$z = \frac{X - \bar{X}}{s} = \frac{65 - 50}{10} = 1.5$$

For history the z score is

$$z = \frac{30 - 25}{5} = 1.0$$

Since the z score for calculus is larger, her relative position in the calculus class is higher than her relative position in the history class.

EXAMPLE 3-28 Test Scores

Find the z score for each test, and state which is higher.

Test A	$X = 38$	$\bar{X} = 40$	$s = 5$
Test B	$X = 94$	$\bar{X} = 100$	$s = 10$

SOLUTION

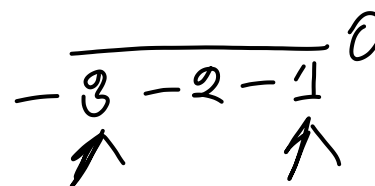
For test A,

$$z = \frac{X - \bar{X}}{s} = \frac{38 - 40}{5} = -0.4$$

For test B,

$$z = \frac{94 - 100}{10} = -0.6$$

The score for test A is relatively higher than the score for test B.



Percentiles

Percentiles are position measures used in educational and health-related fields to indicate the position of an individual in a group.

Percentiles divide the data set into 100 equal groups.

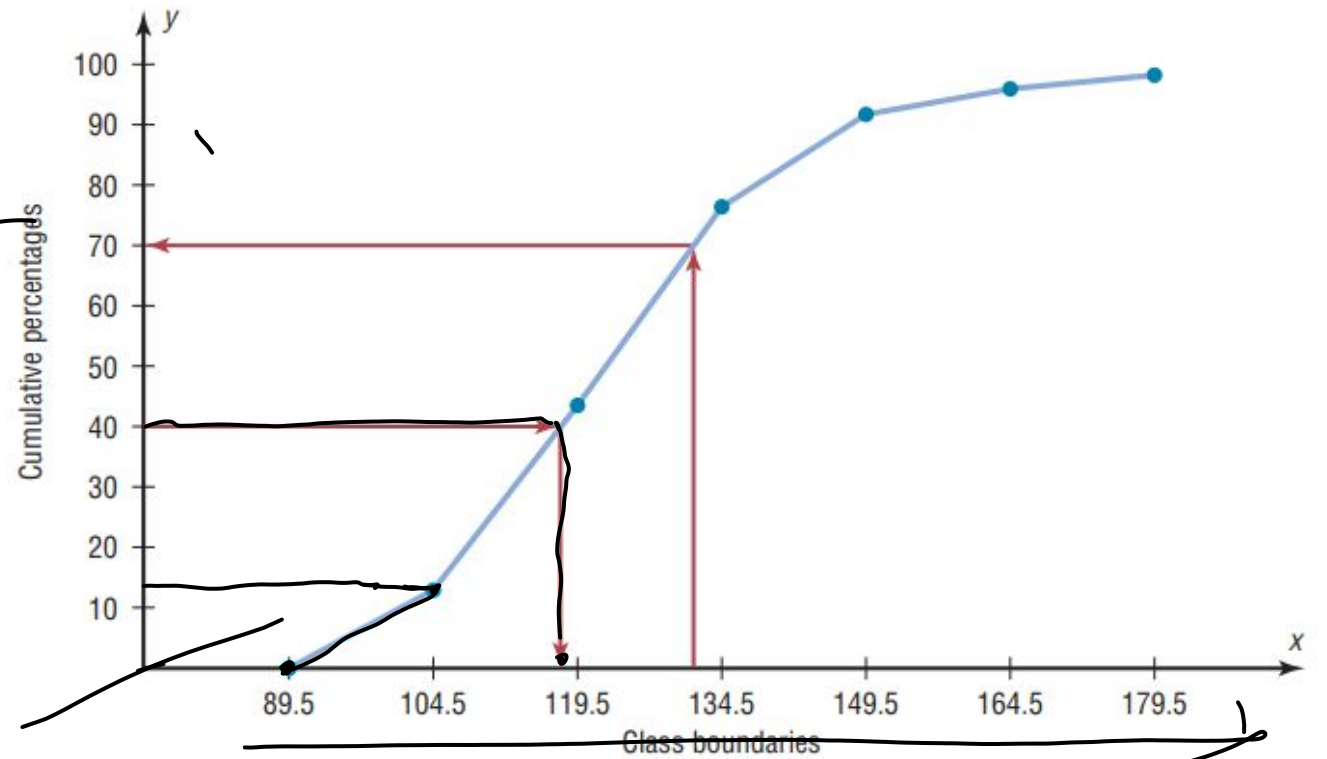
EXAMPLE 3–29 Systolic Blood Pressure

The frequency distribution for the systolic blood pressure readings (in millimeters of mercury, mm Hg) of 200 randomly selected college students is shown here. Construct a percentile graph.

A Class boundaries	B Frequency	C Cumulative frequency	D Cumulative percent
89.5–104.5	24		
104.5–119.5	62		
119.5–134.5	72		
134.5–149.5	26		
149.5–164.5	12		
164.5–179.5	4		
	200		

SOLUTION

A Class boundaries	B Frequency	C Cumulative frequency	D Cumulative percent
89.5-104.5	24	24	12
104.5-119.5	62	86	43
119.5-134.5	72	158	79
134.5-149.5	26	184	92
149.5-164.5	12	196	98
164.5-179.5	4	200	100



Percentile Formula

The percentile corresponding to a given value X is computed by using the following formula:

$$\text{Percentile} = \frac{(\text{number of values below } X) + 0.5}{\text{total number of values}} \cdot 100$$

111
1

12
X

500

EXAMPLE 3-30 Test Scores

A teacher gives a 20-point test to 10 students. The scores are shown here. Find the percentile rank of a score of 12.

18, 15, 12, 6, 8, 2, 3, 5, 20, 10

SOLUTION

Arrange the data in order from lowest to highest.

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

Then substitute into the formula.

$$\text{Percentile} = \frac{(\text{number of values below } X) + 0.5}{\text{total number of values}} \cdot 100$$

Since there are six values below a score of 12, the solution is

$$\text{Percentile} = \frac{6 + 0.5}{10} \cdot 100 = \underline{65\text{th percentile}}$$

Thus, a student whose score was 12 did better than 65% of the class.

EXAMPLE 3-31 Test Scores

Using the data in Example 3–30, find the percentile rank for a score of 6.

Ans: 35th percentile. Do yourself : Page 154

EXAMPLE 3-32 Test Scores

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Using the scores in Example 3-30, find the value corresponding to the 25th percentile.

SOLUTION

Step 1 Arrange the data in order from lowest to highest.

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

Step 2 Compute

$$c = \frac{n \cdot p}{100}$$

where n = total number of values

p = percentile

Thus,

$$c = \frac{10 \cdot 25}{100} = 2.5$$

Step 3 Since c is not a whole number, round it up to the next whole number; in this case, $c = 3$. Start at the lowest value and count over to the third value, which is 5. Hence, the value 5 corresponds to the 25th percentile.

EXAMPLE 3-33

Slide 5

Using the data set in Example 3-30, find the value that corresponds to the 60th percentile.

SOLUTION

Step 1 Arrange the data in order from lowest to highest.

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

Step 2 Substitute in the formula.

$$c = \frac{n \cdot p}{100} = \frac{10 \cdot 60}{100} = 6$$

Step 3 Since c is a whole number, use the value halfway between the c and $c + 1$ values when counting up from the lowest value—in this case, the 6th and 7th values.

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

6th value 7th value

The value halfway between 10 and 12 is 11. Find it by adding the two values and dividing by 2.

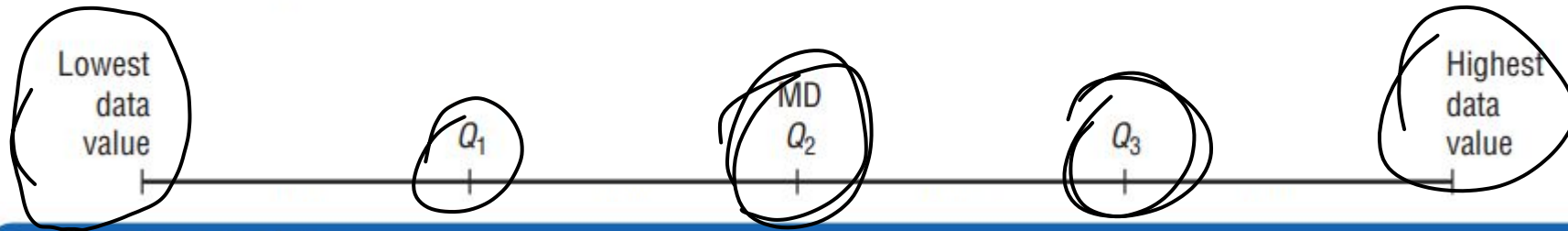
$$\frac{10 + 12}{2} = 11$$

Hence, 11 corresponds to the 60th percentile. Anyone scoring 11 would have done better than 60% of the class.

Quartiles and Deciles

Quartiles divide the distribution into four equal groups, denoted by Q_1 , Q_2 , Q_3 .

Note that Q_1 is the same as the 25th percentile; Q_2 is the same as the 50th percentile, or the median; Q_3 corresponds to the 75th percentile, as shown:



Procedure Table

Finding Data Values Corresponding to Q_1 , Q_2 , and Q_3

- | | |
|---------------|--|
| Step 1 | Arrange the data in order from lowest to highest. |
| Step 2 | Find the median of the data values. This is the value for Q_2 . |
| Step 3 | Find the median of the data values that fall below Q_2 . This is the value for Q_1 . |
| Step 4 | Find the median of the data values that fall above Q_2 . This is the value for Q_3 . |

EXAMPLE 3-34

Find Q_1 , Q_2 , and Q_3 for the data set 15, 13, 6, 5, 12, 50, 22, 18.

SOLUTION

Step 1 Arrange the data in order from lowest to highest.

Step 2 Find the median (Q_2).

5, 6, 12, 13, 15, 18, 22, 50
↑
MD

5, 6, 12, 13, 15, 18, 22, 50

↑
MD

$Q_2 = \frac{13 + 15}{2} = 14$

Step 3 Find the median of the data values less than 14.

5, 6, 12
↑
 Q_1

5, 6, 12, 13

↑
 Q_1

$Q_1 = \frac{6 + 12}{2} = 9$

So Q_1 is 9.

Step 4 Find the median of the data values greater than 14.

15, 18, 22, 50

↑
 Q_3

$Q_3 = \frac{18 + 22}{2} = 20$

Here Q_3 is 20. Hence, $Q_1 = 9$, $Q_2 = 14$, and $Q_3 = 20$.

The **interquartile range (IQR)** is the difference between the third and first quartiles.

$IQR = Q_3 - Q_1$

EXAMPLE 3-35

Find the interquartile range for the data set in Example 3-34.

SOLUTION

$IQR = Q_3 - Q_1 = 20 - 9 = 11$

Outliers

A data set should be checked for extremely high or extremely low values. These values are called *outliers*.

An **outlier** is an extremely high or an extremely low data value when compared with the rest of the data values.

EXAMPLE 3-36 Outliers

Check the following data set for outliers.

{ 5, 6, 12, 13, 15, 18, 22, 50 }

SOLUTION

The data value 50 is extremely suspect. These are the steps in checking for an outlier.

Step 1 Find Q_1 and Q_3 . This was done in Example 3-34; Q_1 is 9 and Q_3 is 20.

Step 2 Find the interquartile range (IQR), which is $Q_3 - Q_1$.

$$\text{IQR} = Q_3 - Q_1 = 20 - 9 = 11$$

Step 3 Multiply this value by 1.5.

$$1.5(11) = 16.5$$

Step 4 Subtract the value obtained in step 3 from Q_1 , and add the value obtained in step 3 to Q_3 .

$$9 - 16.5 = -7.5 \quad \text{and} \quad 20 + 16.5 = 36.5$$

Step 5 Check the data set for any data values that fall outside the interval from -7.5 to 36.5 . The value 50 is outside this interval; hence, it can be considered an outlier.

$$Q_2 = \frac{13 + 15}{2} = 14$$

Procedure Table

Procedure for Identifying Outliers

- | | |
|---------------|---|
| Step 1 | Arrange the data in order from lowest to highest and find Q_1 and Q_3 . |
| Step 2 | Find the interquartile range: $\text{IQR} = Q_3 - Q_1$. |
| Step 3 | Multiply the IQR by 1.5. |
| Step 4 | Subtract the value obtained in step 3 from Q_1 and add the value obtained in step 3 to Q_3 . |
| Step 5 | Check the data set for any data value that is smaller than $Q_1 - 1.5(\text{IQR})$ or larger than $Q_3 + 1.5(\text{IQR})$. |

lower value

$$Q_1 - (\text{IQR} \times 1.5) =$$

$$Q_3 + (\text{IQR} \times 1.5) =$$

↑
upper value

Try yourself:

- 22. Test Scores** Find the percentile rank for each test score in the data set.

12, 28, 35, 42, 47, 49, 50

What value corresponds to the 60th percentile?

- 29.** Check each data set for outliers.

a. 16, 18, 22, 19, 3, 21, 17, 20

b. 24, 32, 54, 31, 16, 18, 19, 14, 17, 20

- 21. Average Weekly Earnings** The average weekly earnings in dollars for various industries are listed below. Find the percentile rank of each value.

804 736 659 489 777 623 597 524 228

For the same data, what value corresponds to the 40th percentile?

- 25. Gasoline Taxes** A random selection of state gasoline taxes per gallon is given below. Find the first and third quartile values for the data.

16 18 35.3 25 23.5 27.1 32.5 16 22
17.5 19 29.5 7.5 12

3–4 Exploratory Data Analysis

The Five-Number Summary and Boxplots

A **boxplot** can be used to graphically represent the data set. These plots involve five specific values:

1. The lowest value of the data set (i.e., minimum)
2. Q_1
3. The median
4. Q_3
5. The highest value of the data set (i.e., maximum)

These values are called a **five-number summary** of the data set.

A **boxplot** is a graph of a data set obtained by drawing a horizontal line from the minimum data value to Q_1 , drawing a horizontal line from Q_3 to the maximum data value, and drawing a box whose vertical sides pass through Q_1 and Q_3 with a vertical line inside the box passing through the median or Q_2 .

EXAMPLE 3-37 Number of Meteorites Found

The number of meteorites found in 10 states of the United States is 89, 47, 164, 296, 30, 215, 138, 78, 48, 39. Construct a boxplot for the data.

Source: Natural History Museum.

SOLUTION

Step 1 Find the five-number summary for the data.

Arrange the data in order:

30, 39, 47, 48, 78, 89, 138, 164, 215, 296

Find the median.

30, 39, 47, 48, 78, 89, 138, 164, 215, 296

Median

Q_2 Median = $\frac{78 + 89}{2} = 83.5$

Find Q_1 .

30, 39, 47, 48, 78

Q_1

Find Q_3 .

89, 138, 164, 215, 296

Q_3

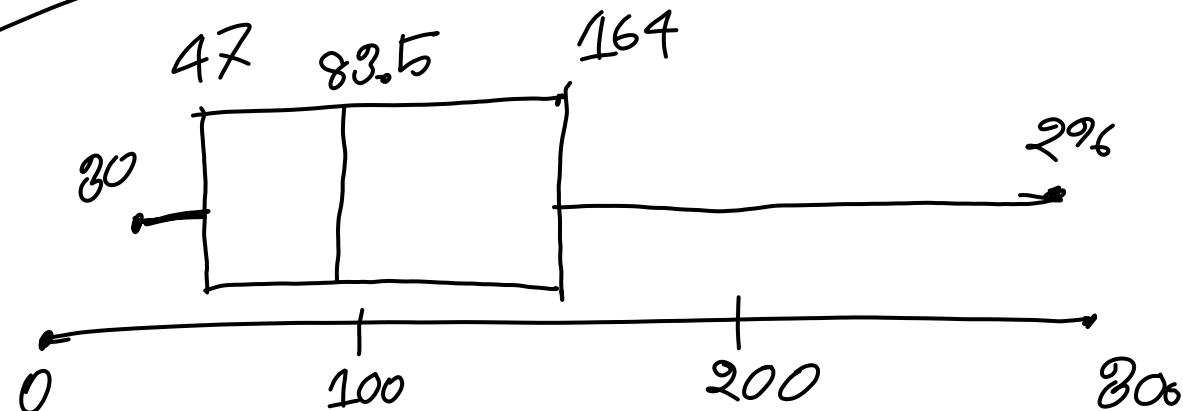
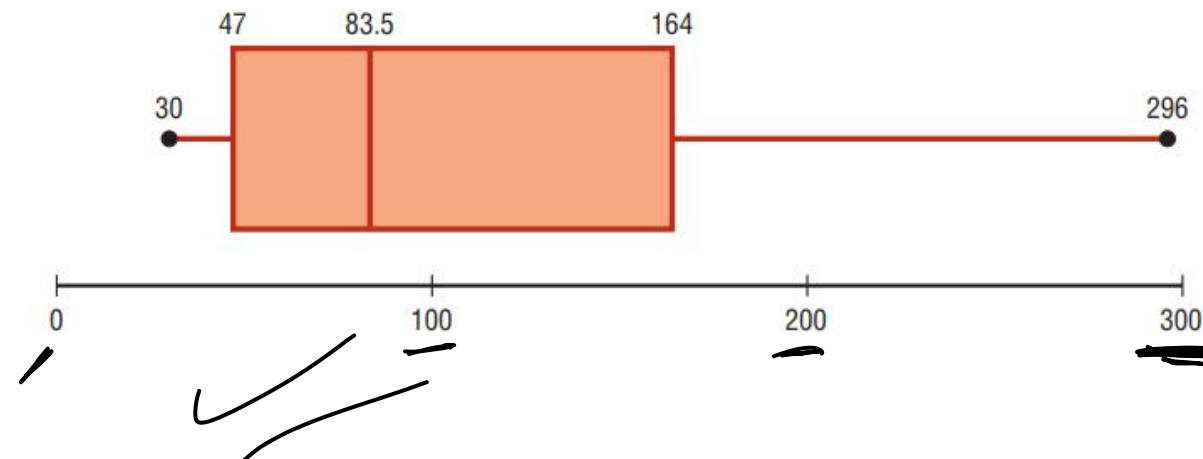


FIGURE 3-7 Boxplot for Example 3-37



Information Obtained from a Boxplot

- 1. a. If the median is near the center of the box, the distribution is approximately symmetric.
b. If the median falls to the left of the center of the box, the distribution is positively skewed.
c. If the median falls to the right of the center, the distribution is negatively skewed.
- 2. a. If the lines are about the same length, the distribution is approximately symmetric.
b. If the right line is larger than the left line, the distribution is positively skewed.
c. If the left line is larger than the right line, the distribution is negatively skewed.

EXAMPLE 3-38 Sodium Content of Cheese

A dietitian is interested in comparing the sodium content of real cheese with the sodium content of a cheese substitute. The data for two random samples are shown. Compare the distributions, using boxplots.

Real cheese				Cheese substitute			
310	420	45	40	270	180	250	290
220	240	180	90	130	260	340	310

SOLUTION

Step 1 Find the five-number summary for each data set. For real cheese

40 45 90 180 220 240 310 420

 ↑ ↑ ↑

Q_1 MD Q_3

$Q_1 = \frac{45 + 90}{2} = 67.5$ MD = $\frac{180 + 220}{2} = 200$

$Q_3 = \frac{240 + 310}{2} = 275$

For cheese substitute

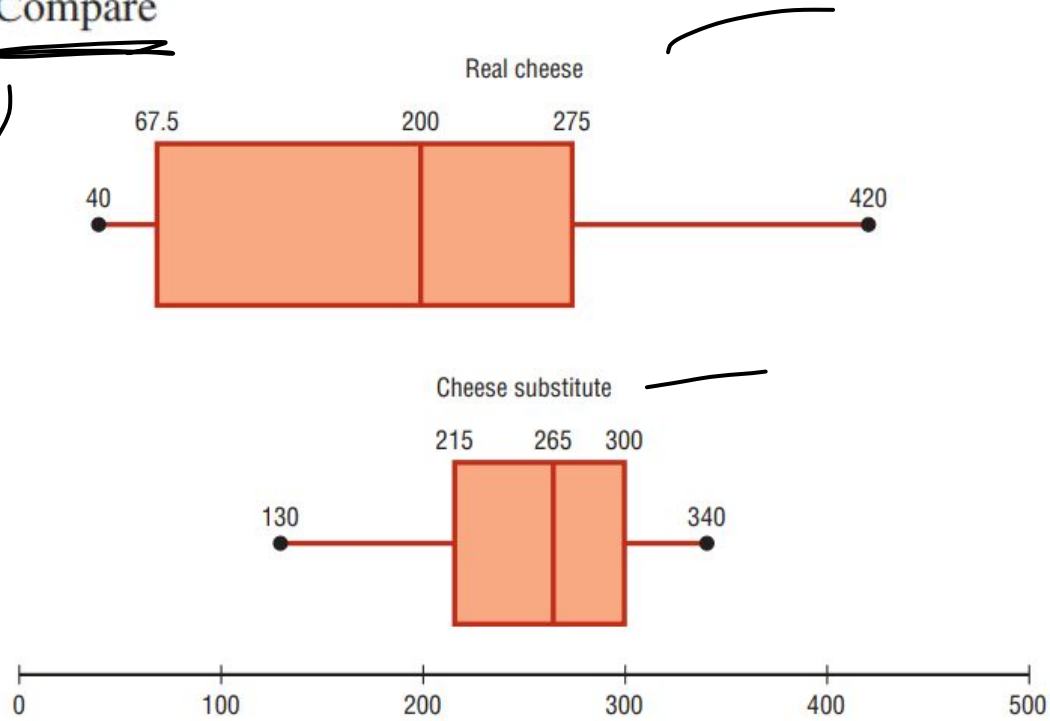
130 180 250 260 270 290 310 340

 ↑ ↑ ↑

Q_1 MD Q_3

$Q_1 = \frac{180 + 250}{2} = 215$ MD = $\frac{260 + 270}{2} = 265$

$Q_3 = \frac{290 + 310}{2} = 300$



Try yourself:

For Exercises 1–6, identify the five-number summary and find the interquartile range.

1. 8, 12, 32, 6, 27, 19, 54

2. 19, 16, 48, 22, 7

- 11. Earned Run Average** Construct a boxplot for the following data and comment on the shape of the distribution representing the number of games pitched by major league baseball's earned run average (ERA) leaders for the past few years.

30 34 29 30 34 29 31 33 34 27 30 27 34 32

Source: *World Almanac*.

- 16. Size of Dams** These data represent the volumes in cubic yards of the largest dams in the United States and in South America. Construct a boxplot of the data for each region and compare the distributions.

United States	South America
125,628	311,539
92,000	274,026
78,008	105,944
77,700	102,014
66,500	56,242
62,850	46,563
52,435	
50,000	

Source: *New York Times Almanac*.